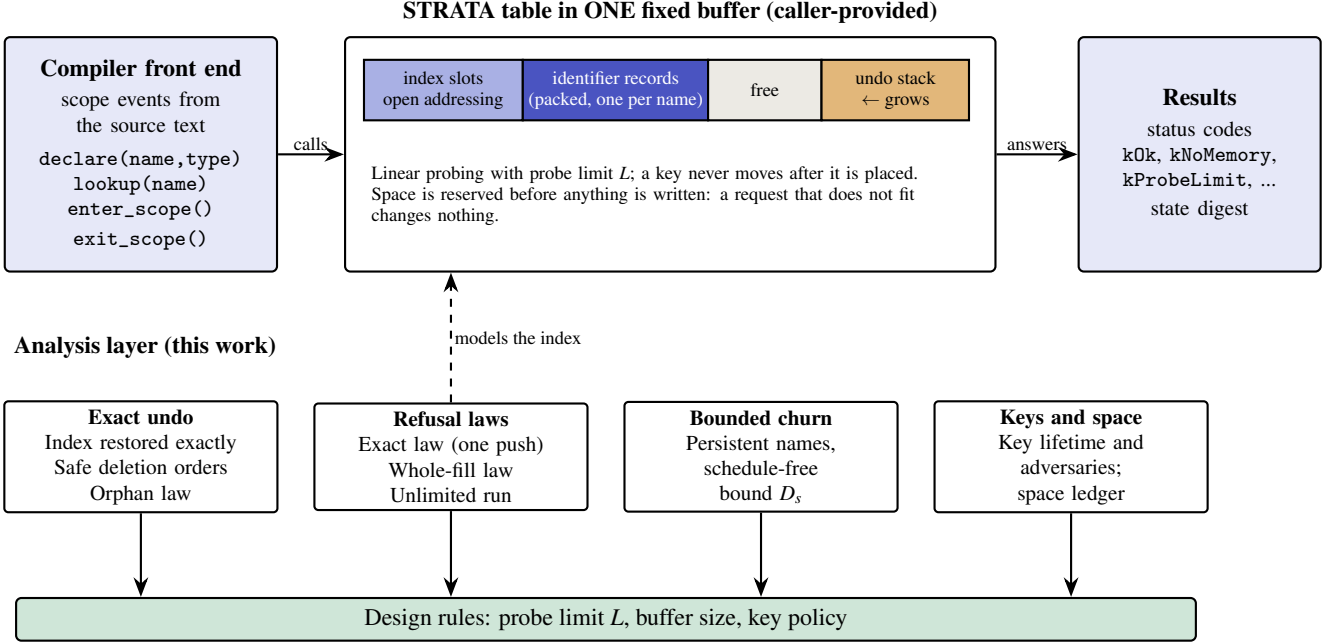

STRATA table in ONE fixed buffer (caller-provided)
Compiler front end
scope events from the source text
declare(name,type)
lookup(name)
enter_scope()
exit_scope()
calls
index slots open addressing
identifier records (packed, one per name)
free
undo stack ← grows
Linear probing with probe limit L; a key never moves after it is placed. Space is reserved before anything is written: a request that does not fit changes nothing.
answers
Results
status codes
kOk, kNoMemory, kProbeLimit, ...
state digest
models the index
Analysis layer (this work)
Exact undo
Index restored exactly
Safe deletion orders
Orphan law
Refusal laws
Exact law (one push)
Whole-fill law
Unlimited run
Bounded churn
Persistent names, schedule-free bound D_s
Keys and space
Key lifetime and adversaries; space ledger
Design rules: probe limit L, buffer size, key policy